

## Day-3 HW

### ☒ Recursion Practice

1. Sum of Digits of a Number
2. Number of Steps to Reduce a Number to Zero (LeetCode 1342)
3. Power of Two (LeetCode 231)
4. Fibonacci Number (LeetCode 509)

---

## 1) Sum of Digits of a Number

### Question

Find the sum of digits of a number using recursion.

Example:

Input: 1234

Output: 10

### C++ Solution

```
int sumDigits(int n) {  
    if (n == 0) return 0;  
    return n % 10 + sumDigits(n / 10);  
}
```

### Java Solution

```
static int sumDigits(int n) {  
    if (n == 0) return 0;  
    return n % 10 + sumDigits(n / 10);  
}
```

### Python Solution

```
def sumDigits(n):  
    if n == 0:  
        return 0  
    return n % 10 + sumDigits(n // 10)
```

---

## 2) Number of Steps to Reduce a Number to Zero (LeetCode 1342)

### Question

If number is even divide by 2, else subtract 1 until it becomes zero.

### C++ Solution

```
int numberOfSteps(int num) {  
    if (num == 0) return 0;  
    if (num % 2 == 0)  
        return 1 + numberOfSteps(num / 2);  
    return 1 + numberOfSteps(num - 1);  
}
```

### Java Solution

```
static int numberOfSteps(int num) {  
    if (num == 0) return 0;  
    if (num % 2 == 0)  
        return 1 + numberOfSteps(num / 2);  
    return 1 + numberOfSteps(num - 1);  
}
```

### Python Solution

```
def numberOfSteps(num):  
    if num == 0:  
        return 0  
    if num % 2 == 0:  
        return 1 + numberOfSteps(num // 2)  
    return 1 + numberOfSteps(num - 1)
```

---

## 3) Power of Two (LeetCode 231)

### Question

Check whether a number is power of two.

### C++ Solution

```
bool isPowerOfTwo(int n) {  
    if (n == 1) return true;  
    if (n <= 0 || n % 2 != 0) return false;  
    return isPowerOfTwo(n / 2);  
}
```

### Java Solution

```
static boolean isPowerOfTwo(int n) {  
    if (n == 1) return true;  
    if (n <= 0 || n % 2 != 0) return false;  
    return isPowerOfTwo(n / 2);  
}
```

### Python Solution

```
def isPowerOfTwo(n):  
    if n == 1:  
        return True  
    if n <= 0 or n % 2 != 0:  
        return False  
    return isPowerOfTwo(n // 2)
```

---

## 4) Fibonacci Number (LeetCode 509)

### Question

Find nth Fibonacci number.

### C++ Solution

```
int fib(int n) {  
    if (n <= 1) return n;  
    return fib(n - 1) + fib(n - 2);  
}
```

### Java Solution

```
static int fib(int n) {  
    if (n <= 1) return n;  
    return fib(n - 1) + fib(n - 2);  
}
```

### Python Solution

```
def fib(n):  
    if n <= 1:  
        return n  
    return fib(n - 1) + fib(n - 2)
```